

Movie & Streaming Content Analytics

SQL Analysis & Business Queries

1. Purpose

This document contains the SQL analysis used to explore the movie and streaming-content dataset. The queries focus on ratings, popularity, audience votes, genres, languages, and release-year trends that support the Power BI dashboard.

2. Dataset Structure

The analysis assumes the dataset is loaded into a PostgreSQL table named **movies**. The main fields used are release date, title, overview, popularity, vote count, vote average, original language, genre, and poster URL.

3. Create Analysis Table

```
CREATE TABLE movies (  
    release_date DATE,  
    title TEXT,  
    overview TEXT,  
    popularity NUMERIC,  
    vote_count INTEGER,  
    vote_average NUMERIC,  
    original_language TEXT,  
    genre TEXT,  
    poster_url TEXT  
);
```

4. Basic Dataset Checks

```
SELECT COUNT(*) AS total_records  
FROM movies;  
  
SELECT  
    ROUND(AVG(vote_average), 2) AS avg_imdb_rating,  
    ROUND(AVG(popularity), 2) AS avg_popularity,  
    ROUND(AVG(vote_count), 2) AS avg_vote_count  
FROM movies;
```

5. High-Rated Titles

```
SELECT title, vote_average, vote_count, popularity  
FROM movies  
WHERE vote_average >= 8  
    AND vote_count >= 100  
ORDER BY vote_average DESC, vote_count DESC;
```

6. Average Rating by Genre

```
SELECT  
    genre,  
    ROUND(AVG(vote_average), 2) AS avg_rating,  
    COUNT(*) AS content_count,  
    ROUND(AVG(vote_count), 0) AS avg_votes  
FROM movies  
WHERE genre IS NOT NULL  
GROUP BY genre  
ORDER BY avg_rating DESC;
```

7. Average Popularity by Genre

```
SELECT  
    genre,  
    COUNT(*) AS content_count,  
    ROUND(AVG(popularity), 2) AS avg_popularity  
FROM movies  
WHERE genre IS NOT NULL  
GROUP BY genre  
ORDER BY avg_popularity DESC;
```

8. Popularity by Release Year

```
SELECT  
    EXTRACT(YEAR FROM release_date)::INT AS release_year,  
    COUNT(*) AS content_count,  
    ROUND(AVG(popularity), 2) AS avg_popularity  
FROM movies  
WHERE release_date IS NOT NULL  
GROUP BY release_year  
ORDER BY release_year;
```

9. Content Growth by Year

```
SELECT  
    EXTRACT(YEAR FROM release_date)::INT AS release_year,  
    COUNT(*) AS content_count  
FROM movies  
WHERE release_date IS NOT NULL  
GROUP BY release_year  
ORDER BY release_year;
```

10. Language Distribution

```
SELECT
    original_language,
    COUNT(*) AS content_count
FROM movies
WHERE original_language IS NOT NULL
GROUP BY original_language
ORDER BY content_count DESC;
```

11. Vote Count vs Rating

```
SELECT
    title, vote_count, vote_average, popularity
FROM movies
WHERE vote_count IS NOT NULL
    AND vote_average IS NOT NULL
ORDER BY vote_count DESC;
```

12. Top Titles by Popularity

```
SELECT
    title, popularity, vote_average, vote_count, genre
FROM movies
ORDER BY popularity DESC
LIMIT 10;
```

13. Top-Rated Titles

```
SELECT
    title, vote_average, vote_count, popularity, genre
FROM movies
WHERE vote_count >= 100
ORDER BY vote_average DESC, vote_count DESC
LIMIT 10;
```

14. Genre and Year Analysis

```
SELECT
    genre,
    EXTRACT(YEAR FROM release_date)::INT AS release_year,
    COUNT(*) AS content_count,
    ROUND(AVG(popularity), 2) AS avg_popularity,
    ROUND(AVG(vote_average), 2) AS avg_rating
FROM movies
WHERE genre IS NOT NULL
    AND release_date IS NOT NULL
GROUP BY genre, release_year
ORDER BY release_year, avg_popularity DESC;
```

15. Dashboard-Oriented Summary

These queries support measures for total content, average rating, average popularity, average votes, high-rated titles, genre comparisons, language distribution, and release-year trends. The resulting data can be connected to Power BI for interactive filtering and visualization.

16. Technologies

Database: PostgreSQL **Query Language:** SQL **Visualization:** Power BI

17. Note

The queries are written for PostgreSQL using the field names represented in the project dataset. If the imported table uses different column names, update the column references before execution.